A Systematic Approach to Intraday Forex Trading

Design, Backtesting, and Robustness Validation of a Multi-Timeframe Momentum and Structural Breakout Strategy

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Quantitative Trading System — Backtesting and Validation Pipeline

Abstract

This paper covers the design, implementation, and validation of a systematic intraday trend-following strategy across three major currency pairs — EURUSD, GBPUSD, and USDJPY — from January 2011 through April 2026. The strategy combines a multi-timeframe directional filter with short-term price-imbalance zones on the five-minute series as limit-order entry regions, with structure-referenced stops and a two-stage scaled exit. The core thesis is that multi-timeframe directional persistence and intra-session liquidity-gap rebalancing interact to produce a conditional structural edge in liquid FX pairs.

A Python-based backtesting pipeline evaluates performance under three execution-cost regimes, including a worst-case stress test at 2× estimated spread and slippage. Portfolio simulation runs shared capital across all three instruments with continuous open-risk accounting and a 5.5% portfolio cap. Under realistic execution costs, the three-pair portfolio returned +136.71% over the 15-year test period, with a maximum drawdown of -5.36% and an annualized Sharpe of 1.60. Under worst-case execution, those numbers compress to +62.97%, -8.74% drawdown, and a Sharpe of 0.87.

Robustness testing — walk-forward validation, rolling window analysis, parameter sensitivity, and a 1,000-iteration bootstrap Monte Carlo — points away from overfitting or favorable trade sequencing as explanations for the results. The t-statistic on per-trade returns exceeds conventional significance thresholds, and every bootstrap resample finishes above starting capital. The strategy is best characterized as a regime-conditional structural edge: strongest during sustained directional phases, weakest when macro shocks or structural disruption dominate. The results support a positive expected-value interpretation, with regime dependency and execution-modeling constraints as the primary forward-looking risks.

1. Introduction

This paper documents the development of a fully systematic intraday trend-following strategy for spot forex. The motivation is straightforward: directional continuations following multi-timeframe structural alignment in highly liquid currency pairs appear to be an exploitable regularity, conditional on specific intra-session liquidity behavior. The goal was to translate that observation into a precise, rule-based system with quantifiable expected value, explicit execution-cost modeling, and a validation framework suitable for assessing live deployment readiness.

The design requirements were fourfold. Every entry, exit, and risk decision must be fully deterministic — no discretionary override. Execution costs must be modeled explicitly across multiple scenarios, including a stress case that substantially exceeds typical conditions. Performance must be validated on data not used in parameter selection, using both walk-forward and rolling window methods. And the portfolio simulation must reflect shared capital constraints rather than treating each instrument independently.

The result is an eight-module Python pipeline covering data ingestion, signal generation, trade simulation, cost modeling, analytics, walk-forward validation, sensitivity analysis, and portfolio simulation. The sections that follow cover the strategy's theoretical basis, simulation architecture, robustness validation methodology, statistical significance testing, portfolio-level results under two cost scenarios, a benchmark comparison, an explicit limitations analysis, and an edge classification that defines the conditions under which the strategy is expected to perform.

2. Problem Statement

Systematic strategy development requires resolving three evaluation challenges before any performance figures can be taken seriously.

The first is measurement. Without a fixed, rule-based signal definition, there's no way to distinguish genuine positive expected value from selective observation under favorable conditions. Formalizing the strategy in code forces complete specification of every decision, making performance deterministic and auditable.

The second is reproducibility. A fully specified simulation engine applies identical logic to every bar in the historical series, removing variation from changing market conditions or inconsistent execution. With a fixed random seed, results are reproducible across runs and environments.

The third is cost sensitivity. For an intraday strategy with modest per-trade R-multiples, spread and slippage eat a meaningful fraction of expected return. A strategy that looks profitable under zero-cost assumptions can turn negative once realistic friction is applied. Explicit cost modeling and multi-scenario testing are required before any capital goes to work.

This project addresses all three by building a deterministic, fully logged simulation engine that runs on tick-approximate one-minute OHLCV data, with explicit cost modeling and structured performance output across multiple execution scenarios.

3. Strategy Design

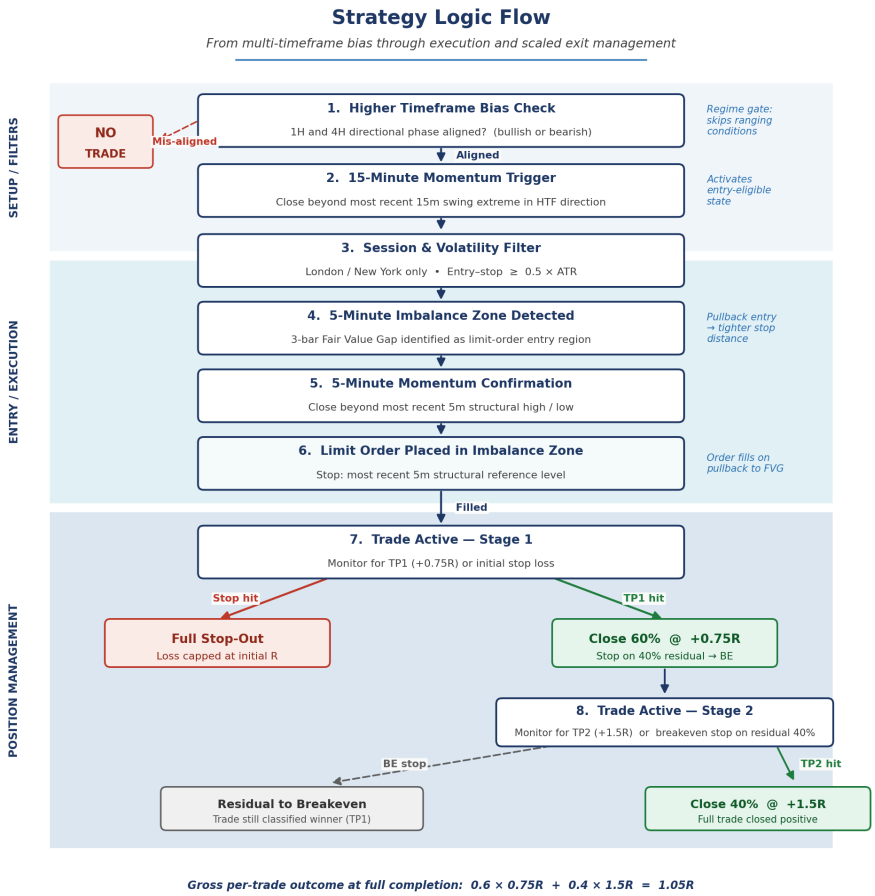

Figure 1. Strategy Logic Flow Diagram

Complete decision tree from higher-timeframe bias through execution and scaled exit management. Each block is a deterministic gate in the entry pipeline; failure of any upstream condition prevents the trade from being evaluated at the next stage.

3.1 Overview

The strategy is a systematic intraday trend-following model running on five-minute execution data with multi-timeframe directional conditioning. Entry triggers on short-term momentum confirmation in the direction of the dominant higher-timeframe bias, using a limit order targeting a defined price-imbalance zone. Risk is managed through a structure-referenced initial stop and a two-stage scaled exit — partial profit is locked in at a near target while the residual position retains directional exposure.

3.2 Structural Reference Points

Structural reference points are local extrema of price. A swing high is the bar whose high is maximal within a symmetric lookback window on both sides; a swing low is the corresponding minimum. Ascending sequences — higher highs and higher lows — define an upward structural phase. Descending sequences define a downward phase.

3.3 Multi-Timeframe Directional Filter

Before any entry is evaluated, both the one-hour and four-hour timeframes must show consistent directional phase alignment. Phase on each timeframe is determined by the most recently confirmed structural sequence — ascending local extrema for a bullish phase, descending for bearish.

This dual-timeframe filter acts as a regime gate. It systematically excludes entries during structural ambiguity or mean-reverting price action, where a momentum-following model has no theoretical basis for positive expected value. Requiring phase agreement across two independent timeframe resolutions reduces entry frequency while meaningfully improving the conditional base rate of directional continuation. If the two timeframes disagree, no trade is evaluated.

3.4 Intermediate-Timeframe Momentum Trigger

With higher-timeframe phase confirmed, the strategy monitors the fifteen-minute series for a momentum trigger: a close above the most recent 15m swing high on the long side, or a close below the most recent 15m swing low on the short side. This confirms that intraday momentum has aligned with the dominant multi-timeframe bias and activates the execution-level entry scan.

3.5 Intra-Session Price-Imbalance Zones

With the intermediate momentum trigger active, the strategy identifies intra-session price-imbalance zones on the five-minute series. An imbalance zone is a three-bar pattern where sequential auctions leave a gap in bid-ask overlap: the high of the first bar doesn't reach the low of the third (bullish imbalance), or the low of the first bar doesn't reach the high of the third (bearish imbalance). These zones mark regions where the market transacted rapidly without full two-sided participation — a pocket of structural inefficiency that price tends to revisit before continuing.

The strategy uses these zones as limit-order entry regions, exploiting the mean-reversion tendency back into the imbalance rather than chasing the breakout. This improves average entry relative to the eventual directional move and tightens the stop-to-target distance, which directly improves expected value per trade.

3.6 Execution-Timeframe Momentum Confirmation

Before placing a limit order, the strategy requires a momentum confirmation signal on the five-minute series: a close above the most recent 5m structural high for longs, or below the most recent structural low for shorts. This ensures execution-timeframe momentum is co-directional with the higher-timeframe phase before the order goes in, and filters out entries that would otherwise trigger during passive price drift with no genuine directional momentum behind them.

3.7 Limit-Order Entry via Mean-Reversion to Imbalance Zone

Entry isn't placed at the momentum confirmation bar. Instead, a limit order is set within the imbalance zone, targeting the mean-reversion of price back into the gap following the breakout. This pullback mechanism tightens the stop-to-target distance by placing entry closer to the structural invalidation level while keeping the same target. Entering within the zone rather than at the breakout origin reduces average fill price on longs and increases it on shorts — both directionally favorable to expected value. The limit order activates when the five-minute close re-enters the imbalance zone boundaries after momentum confirmation.

3.8 Reference-Level Stop Placement

The initial stop sits at the most recent five-minute structural reference level opposite to the trade direction — the most recent structural low for longs, the most recent structural high for shorts, derived from a rolling lookback over recent bars.

A move back through that level indicates the trade's momentum premise has been invalidated: multi-timeframe phase alignment with execution-level momentum confirmation no longer holds. Structure-referencing the stop means it adapts to prevailing volatility — wider in high-volatility regimes, tighter in low-volatility ones — which reflects a market-defined invalidation criterion rather than an arbitrary fixed distance. If no structural level falls within a minimum distance threshold of at least 0.5× ATR, the stop defaults to a fixed ATR-multiple distance. This prevents trades from running with stops so narrow they'd be triggered by normal execution noise rather than genuine structural failure.

3.9 Scaled Exit Model

The exit model runs in two stages. At TP1, placed at 0.75R from entry, 60% of the position closes and the stop on the remaining 40% moves to entry. TP2 sits at 1.5R, where the residual closes in full.

The two-stage structure is calibrated to MFE analysis: most profitable trades reach at least 0.75R before retracing, which makes TP1 achievable without requiring large directional continuation. Moving the stop to entry after TP1 eliminates further capital at risk on the residual while preserving exposure to the 1.5R outcome. The result is a structurally elevated win rate: any trade reaching TP1 is a winner regardless of what the runner does. The average win reflects the weighted combination of TP1-only and full TP1+TP2 outcomes, and the average loss is bounded by the initial risk on the 60% partial that stops out before TP1.

4. System Architecture and Code Implementation

4.1 Overview

The project is implemented in Python across eight modules, each with a distinct responsibility. Components are designed to be tested and replaced independently — the portfolio simulation consumes results from the per-pair engine without any knowledge of its internal logic.

4.2 trailing.py — Core Strategy Engine

trailing.py is the central simulation engine, implementing all strategy logic: signal detection, entry conditions, stop and target management, and trade logging. The function run_backtest() accepts a dictionary of resampled OHLCV DataFrames and execution-cost parameters, and simulates every bar of the five-minute series in chronological order.

Core responsibilities include higher-timeframe bias evaluation at each bar using precomputed HTF index maps, intermediate-timeframe momentum trigger detection, five-minute price-imbalance zone identification, execution-level momentum confirmation, limit-order entry activation, structure-referenced stop computation, and position management through the scaled exit model. The engine produces three outputs: a trade log DataFrame with one row per trade, an exit-event DataFrame recording each individual leg closure (TP1, TP2, stop, or end-of-data), and an equity-curve DataFrame tracking account value over time.

Several filters layer on top of the core signal logic. An ATR-based volatility filter requires the trade's R-distance to meet a minimum threshold, preventing entries where the structure stop is close enough that normal noise would trigger it. A session filter restricts entries to London and New York hours, where liquidity is higher and spread costs are proportionally smaller. A break-of-structure quality check requires that the five-minute BOS represents a genuine structural shift rather than a trivial bar-level move.

Performance was a real concern given that a five-minute bar series spanning 15 years runs into hundreds of thousands of bars. Replacing per-bar computations — ATR, structural reference levels, momentum confirmation swing references, imbalance-zone detection, and HTF index lookups — with precomputed arrays and vectorized operations produced roughly a 19× speedup, cutting a single-pair full backtest from over ten minutes to under forty seconds.

4.3 sim_costs.py — Execution Cost Modeling

sim_costs.py defines the cost model applied to every entry and exit across three scenarios: ideal (zero cost), realistic (standard spreads and slippage), and worst-case (double both). Spread and slippage parameters are derived from publicly available institutional broker data during active sessions. Entry fills are adjusted by spread; exits by a normally distributed slippage draw. Cost impact varies trade-to-trade rather than applying a constant penalty.

4.4 data_loader.py — Data Loading and Resampling

data_loader.py accepts a path to a one-minute OHLCV CSV for a given pair and resamples it into all required timeframes: 5m, 15m, 1H, and 4H, using standard OHLCV aggregation rules. All timeframes are indexed consistently so that at any 5m bar, the corresponding higher-timeframe bar can be looked up without misalignment — critical for preventing look-ahead bias, since the system must only use information available at the close of the current bar.

4.5 analytics.py — Performance Measurement

analytics.py provides post-hoc analysis functions for evaluating completed backtest results. It computes standard performance metrics — Sharpe ratio, maximum drawdown, profit factor, win rate, expectancy, and CAGR — alongside Maximum Favorable Excursion (MFE) and Maximum Adverse Excursion (MAE) on a per-trade basis. MFE measures the largest unrealized gain during a trade's lifetime and calibrates target placement against the empirical distribution of favorable price movement. MAE measures the largest adverse intra-trade excursion and validates whether stops are correctly positioned relative to typical noise levels.

The module also provides a yearly breakdown segmenting performance by calendar year for consistency evaluation across regimes, and a continuation analysis measuring how often price reached various R-multiples and how often it retraced to breakeven after TP1.

4.6 robustness.py — Walk-Forward and Sensitivity Testing

robustness.py implements three forms of robustness testing. Walk-forward validation divides the data into sequential in-sample and out-of-sample windows, testing whether the strategy maintains positive performance on data it never saw during parameter selection. Rolling window testing applies the strategy to fixed-length sub-periods across the full history, producing a distribution of performance metrics rather than a single aggregate number. Parameter sensitivity testing varies key inputs — ATR multiplier, TP ratios, lookback windows — and measures how performance degrades as parameters move away from the baseline configuration.

The purpose is to determine whether the full-period backtest results are genuinely robust, or whether they depend on a specific calibration that happened to fit the historical data well. A strategy whose performance falls apart when parameters are nudged is likely overfit to the historical series.

4.7 validate.py — Efficient Validation Runner

validate.py orchestrates the full validation pipeline. Its core function is avoiding redundant computation — the swing detection pass, data loading, and baseline backtest only need to run once, with results cached and passed to downstream validation routines. Without this coordination, a full robustness suite would recompute the same base quantities dozens of times.

The module precomputes swing points for the full date range and passes them into the parameter robustness tests, which operate on the full date range and benefit from the precomputed data. They are deliberately not passed into walk-forward or rolling window tests, where data is sliced to sub-periods — using full-range swing data there would introduce look-ahead bias.

4.8 portfolio.py — Portfolio Simulation

portfolio.py simulates all three pairs trading simultaneously against a single shared equity account. Rather than running three separate backtests and summing the results, the module replays all trade events chronologically and updates a single equity balance at each event. Per-pair backtests produce entry and exit events with timestamps and PnL values, which are merged into a unified chronological event stream and replayed against shared capital. At each entry, the trade is sized based on current portfolio equity and the configured risk percentage — a scale factor is stored and applied consistently to all subsequent exit events for that trade. When events share a timestamp, full closes are processed before partial closes, and partial closes before new entries, ensuring equity is current before any new position is sized.

Open-risk accounting supports the portfolio cap used in the final configuration. At each entry event, the trade's dollar risk is added to a running total. After TP1, when the stop moves to breakeven, that trade's contribution to open risk drops to zero. At each full close, the remaining risk is removed. Before accepting a new entry, the simulation checks whether adding the new trade's risk would push total open risk above the configured cap as a fraction of portfolio equity — if it would, the trade is rejected. This prevents the simultaneous accumulation of excessive correlated risk across pairs during periods when all three markets are active at once.

4.9 run_portfolio.py — Portfolio Runner

run_portfolio.py is the entry point for the full portfolio simulation. It configures the pairs, data file paths, risk percentages, portfolio cap, starting equity, and cost scenario, then calls the portfolio pipeline. Switching to the worst-case scenario is a single flag that doubles both spread and slippage multipliers and routes output to a separate directory. The file runs directly from the terminal with no command-line arguments required.

5. Backtesting Methodology

5.1 Data

The backtest uses one-minute OHLCV data for EURUSD, GBPUSD, and USDJPY spanning January 2011 through April 2026 — roughly 15 years and 207 completed trades across the three pairs under the realistic execution scenario. Data comes from a local historical dataset aggregated from Dukascopy tick data, a standard reference source for retail-grade forex backtesting. No external feeds or live APIs are used during simulation.

Before backtesting, each one-minute CSV was screened for duplicate timestamps, which were dropped, and extended weekend gaps (Friday close to Sunday open) were excluded from session logic. Bars outside the defined trading sessions were retained in the data but filtered at the signal level — this way HTF structure can still be computed across non-trading hours without those hours generating entries. The one-minute base data resamples in memory at runtime to 5m, 15m, 1H, and 4H timeframes. All plots cover the full January 2011 through April 2026 range and display final equity values at the terminal bar, with no axis truncation applied.

5.2 Simulation Approach

The simulation is event-driven at the bar level: the engine iterates over every five-minute bar in order, applying all strategy logic using only information available at the close of that bar. No future data is referenced at any point. Higher-timeframe values are derived by mapping each five-minute bar index to the corresponding closed higher-timeframe bar via a pre-built index-alignment array.

This design prevents look-ahead bias by construction. The five-minute bar at time T can only reference the one-hour bar that closed before T — never the one that closes after. The simulation is fully deterministic: NumPy's RNG is seeded with a fixed value before each run, so identical inputs always produce identical outputs. Results are reproducible across machines given the same data files and code version.

5.3 Trade Logging

Every trade is recorded with full metadata: entry time and price, direction, original stop price, TP1 and TP2 prices, exit time and price, exit type, PnL in dollars, PnL as a percentage of equity at entry, whether each target was hit, and equity after the trade closes. .

5.4 Portfolio Scaling

The portfolio simulation scales each trade's PnL from the per-pair simulation to portfolio equity using the scale factor derived at entry. This means the portfolio results aren't simply the sum of three independent simulations — they reflect the actual equity available at the time each trade was entered.

6. Cost Modeling and Execution Assumptions

6.1 Motivation

For an intraday strategy with moderate trade frequency, execution costs are a meaningful fraction of each trade's expected value — targets are small multiples of initial risk, so spread and slippage eat into returns quickly. A strategy that looks profitable before costs can turn marginal or negative once realistic friction is applied. Testing across multiple cost scenarios quantifies that degradation and establishes whether the edge holds up under execution uncertainty.

6.2 Ideal Scenario

The ideal scenario applies zero spread and zero slippage, with fill prices matching theoretical signal prices exactly — an upper bound on raw edge before any cost drag.

6.3 Realistic Scenario

The realistic scenario applies spread and slippage estimates based on typical retail broker conditions for each pair during active hours. EURUSD has the tightest spread and lowest slippage given its liquidity depth. USDJPY runs slightly wider. GBPUSD sits between the two. Entry fills are adjusted by the spread; exit fills add a random slippage draw from a normal distribution with the configured standard deviation.

Slippage is parameterized with a mean of $\mu = 0$ pips — no expected directional bias — and per-pair standard deviations of $\sigma = 0.3$ pips for EURUSD, 0.5 pips for USDJPY, and 0.4 pips for GBPUSD. Draws are unclipped, so occasional large slippage events can and do occur, consistent with real-world tail execution. The worst-case scenario doubles each σ value while leaving μ unchanged.

6.4 Worst-Case Scenario

The worst-case scenario doubles both spread and slippage standard deviation across all pairs. It isn't meant to represent expected conditions — it's a stress test against adverse execution: wider spreads, higher slippage, and generally poor fill quality. If the strategy stays viable here, it has real tolerance for execution deterioration.

7. Strategy Refinement

7.1 Initial Development

The baseline signal specification was tested without session filtering, minimum R-distance filtering, or execution-level momentum quality constraints. Raw trade frequency was higher, but risk-adjusted performance was substantially weaker — lower profit factor, higher maximum drawdown. A significant proportion of those trades occurred during the Asian session, where spreads are wider and intraday directional continuation rates are lower.

7.2 Session Liquidity Filter

Restricting entries to the London session (08:00–12:00 UTC) and New York session (13:00–17:00 UTC) reduced trade count while improving quality. These windows represent peak liquidity for all three pairs — spreads are tightest and institutional order flow is most likely to produce sustained directional movement. Excluding the Asian session removed a disproportionate share of structurally weak entries and improved profit factor substantially without touching the core signal logic.

7.3 Minimum R-Distance Filter

The minimum R-distance filter requires that the entry-to-stop distance equal at least a defined fraction of the prevailing ATR. Trades with stops very close to entry are disproportionately sensitive to execution slippage — a fill deviation of even a fraction of a pip can eliminate the trade's risk-reward ratio. The filter removes entries where the stop distance is too narrow to distinguish genuine directional failure from routine execution noise, improving the signal-to-noise ratio of the remaining trade set.

7.4 Momentum Threshold Filter

The momentum threshold filter requires that the execution-timeframe confirmation signal represent a structurally meaningful price move — not a marginal extension of the prior high or low. Without it, the strategy generates entries on trivial structural extensions that don't reflect genuine momentum. Adding this filter reduced false positives in low-momentum, oscillating conditions.

7.5 Exit Mode Selection

The system supports three exit modes: dual_tp (the primary mode described in Section 3.9), single_tp (a single target at a fixed R-multiple), and fixed_sl (stop loss only, no take profit). After testing all three, dual_tp produced the best risk-adjusted performance across both realistic and worst-case cost scenarios. The single-target mode generated higher average wins per trade but a lower win rate, resulting in a worse Sharpe ratio. The fixed stop loss mode held up reasonably well in strong trending conditions but degraded significantly during range-bound periods.

8. Portfolio Construction

8.1 Design Rationale

Treating each pair as a separate equity account understates realized risk — if EURUSD, GBPUSD, and USDJPY enter simultaneously, the single-account trader bears the sum of those positions. A simulation that treats the three equity curves independently will understate drawdown and overstate Sharpe. The portfolio module replays all trade events chronologically against a single shared equity balance, with position sizes scaled to current portfolio equity at each entry rather than a fixed starting balance.

8.1.1 Yearly Breakdown

Annual returns were positive in every complete calendar year in the sample, with variation tracking the strategy's regime dependence. The strongest years — 2017 (+11.1%), 2018 (+12.8%), and 2019 (+12.2%) — coincide with sustained multi-pair directional movement. The weakest consecutive run — 2020 (+2.1%), 2021 (+1.1%), 2022 (+0.2%) — reflects the macro-driven disruption analyzed in Section 15.3. More relevant than the absence of a losing year is that no year breached the 10% drawdown threshold, consistent with portfolio-level risk control remaining operative across diverse market conditions.

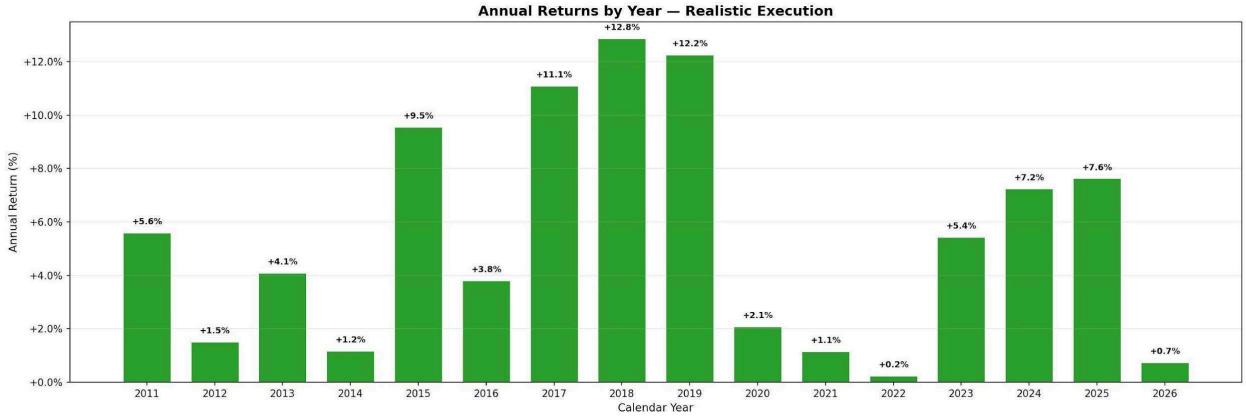

Figure 15. Annual Returns by Year — Realistic Execution

Per-year portfolio returns across the full test period. Returns remain generally consistent, with a visible drawdown during the 2020–2022 macro-disruption period.

8.1.2 Pair Contributions

All three pairs contributed positively to total portfolio PnL, indicating the signal generalizes across instruments rather than being driven by a single pair. EURUSD (+$53,848, 73% win rate) and USDJPY (+$41,535, 72% win rate) produced the largest absolute contributions, consistent with their higher 1.5% per-trade risk allocation. GBPUSD (+$33,644, 81% win rate) contributed less in absolute terms but posted the highest win rate of the three, consistent with its tighter 1.0% allocation. All three maintained profit factors above 2.0 across the full test period.

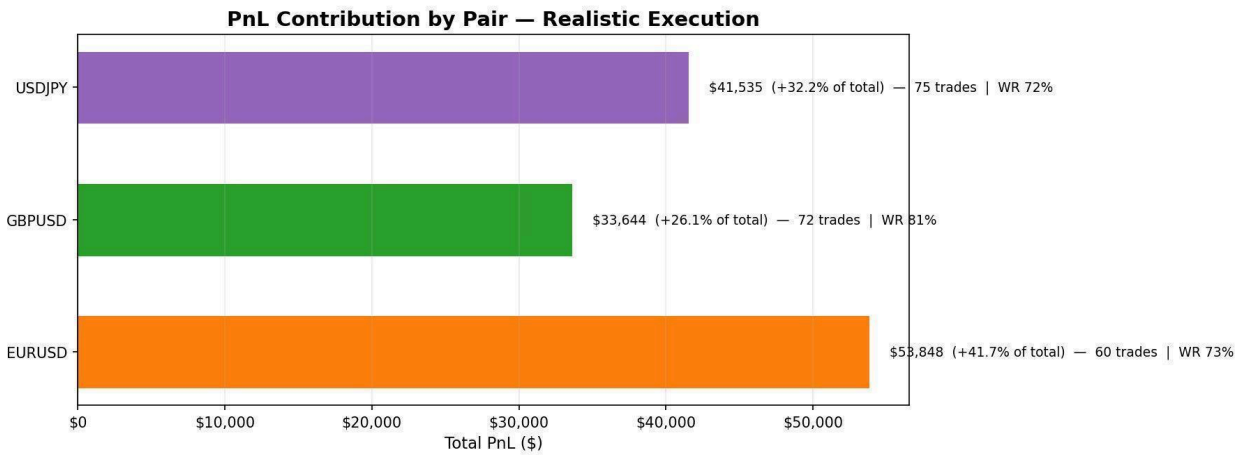

Figure 16. PnL Contribution by Pair — Realistic Execution

Absolute PnL, trade count, and win rate by instrument. EURUSD contributes the most PnL, while GBPUSD shows the highest win rate.

8.1.3 Overlap Analysis

The majority of entries occurred with either no other open position or just one other open simultaneously. All three pairs holding positions at the same time was infrequent, and when it did occur, the 5.5% portfolio cap bounded total concurrent risk. Maximum concurrent open risk stayed within the configured cap throughout the test period.

8.2 Risk Configuration and Portfolio Cap

Per-instrument risk is set at 1.5% per trade for EURUSD and USDJPY, and 1.0% for GBPUSD — reflecting differences in historical signal quality and profit factor across instruments. Total portfolio open risk is capped at 5.5% at any point. Given the strategy's trade frequency and typical intraday hold duration, simultaneous multi-pair exposure is infrequent, so the cap functions as a correlated-risk circuit breaker rather than a routine constraint.

8.3 Dynamic Open-Risk Accounting

Open risk is tracked in dollar terms continuously throughout the simulation. At entry, a trade's initial capital-at-risk is added to the running total. After TP1, when the stop moves to entry, that trade's contribution drops to zero — reflecting the elimination of further downside on the position. At full close, any residual risk is cleared.

8.4 Event Processing Order

When multiple events share a timestamp, the simulation processes full closes first, then partial closes, then new entries — so new positions are always sized on the most current equity.

9. Results

9.1 Realistic Execution Scenario

Under the realistic execution scenario, the three-pair portfolio produced the following results over the January 2011 through April 2026 test period.

Metric	Value
Total Trades	207
Win Rate	77.3%
Total Return	+136.71%
Max Drawdown	-5.36%
Sharpe Ratio	1.600
Profit Factor	2.581
Final Equity	$236,708.03

Starting from $100,000, the portfolio compounded to $236,708 over the test period. The win rate reflects the dual-stage exit structure — any trade reaching TP1 before stop-out is classified as a winner regardless of TP2 outcome. A profit factor of 2.58 means gross profit exceeded gross loss by 2.58:1 under realistic costs. The maximum drawdown of 5.36% occurred in August 2014 and represents the largest peak-to-trough decline over the full test period.

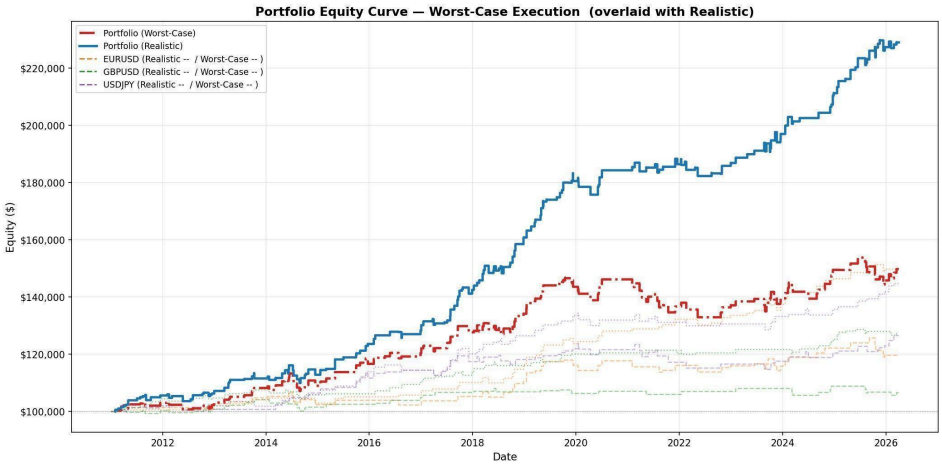

Figure 17. Portfolio Equity Curve — Realistic vs Worst-Case

Equity curves under realistic and worst-case execution, showing steady growth and controlled drawdowns.

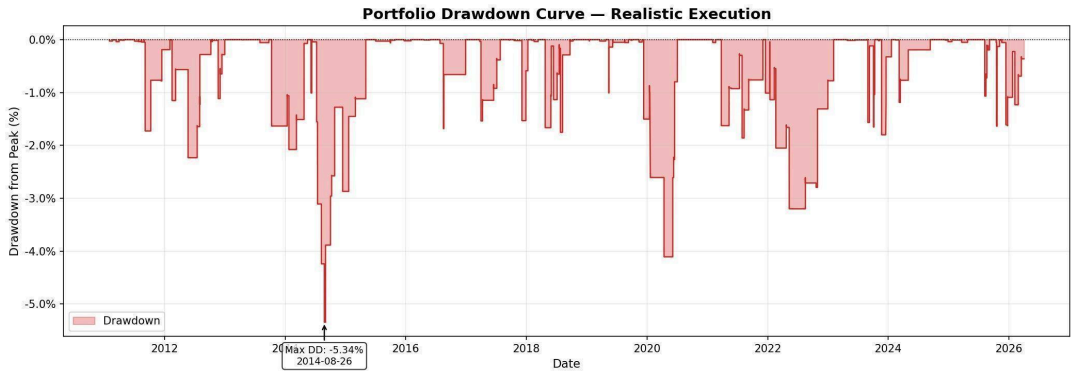

Figure 18. Portfolio Drawdown Over Time

Drawdown over the full test period, showing controlled declines and stable risk management.

The Sharpe ratio is computed on per-trade returns annualized using the observed trade frequency and date range: $S = (\mu\square - r_x) / \sigma\square$, where $\mu\square$ is the annualized mean per-trade return, r_x is the risk-free rate (set to zero for forex), and $\sigma\square$ is the annualized standard deviation of per-trade returns.

9.2 Worst-Case Execution Scenario

Under the worst-case execution scenario — 2× realistic spread and 2× slippage standard deviation applied uniformly across all instruments and periods — the portfolio produced the following results.

Metric	Value
Total Trades	171
Win Rate	71.9%
Total Return	+62.97%
Max Drawdown	-8.74%
Sharpe Ratio	0.867
Profit Factor	1.681
Final Equity	$162,974.63

Trade count fell from 207 to 171 (−17.4%) as wider entry spreads pushed additional trades below the minimum R-distance filter — a form of natural self-correction. Win rate declined 5.4 percentage points, reflecting higher cost drag at TP1. Profit factor compressed from 2.58 to 1.68, retaining positive expected value above 1.0. The Sharpe ratio dropped to 0.867, below the conventional 1.0 threshold — acceptable performance, but below benchmark on a risk-adjusted basis.

The worst-case scenario is a conservative execution floor. Doubled spread and doubled slippage sustained uniformly across 15 years exceeds what would realistically be experienced even under materially degraded conditions. Retaining profitability under these assumptions suggests the edge isn't solely attributable to favorable execution.

10. Comparative Analysis: Realistic vs Worst-Case

Metric	Realistic	Worst-Case	Change
Trades	207	171	-17.4%
Win Rate	77.3%	71.9%	-5.4 pp
Total Return	+136.71%	+62.97%	-73.74 pp
Max Drawdown	-5.36%	-8.74%	-3.38 pp
Sharpe Ratio	1.600	0.867	-0.733
Profit Factor	2.581	1.681	-0.900
Final Equity	$236,708	$162,975	-$73,733

The roughly 74 percentage point return differential between scenarios has two compounding sources: higher per-trade cost drag directly reduces PnL, and the 17.4% reduction in trade count limits compounding opportunities.

The Sharpe decline from 1.60 to 0.867 reflects both lower mean per-trade return and higher outcome dispersion — wider slippage draws increase PnL variance, compressing the ratio of mean to standard deviation. The 3.38 percentage point increase in maximum drawdown reflects wider losses when slippage exceeds the TP1 distance on marginal trades. Both figures stay below 10%, consistent with portfolio-level risk control holding across both scenarios.

Profit factor exceeds 1.0 under both scenarios, with compression from 2.58 to 1.68 quantifying the cost sensitivity — roughly 35% of the gross profit buffer is consumed when execution costs are doubled. The 17.4% trade count reduction under worst-case costs is structurally favorable: as entry costs rise, the most marginal trades fail the minimum R-distance filter and are automatically discarded, partially offsetting per-trade cost degradation.

11. Validation and Robustness Testing

11.1 Walk-Forward Validation (In-Sample / Out-of-Sample)

The full 15-year series was partitioned into sequential non-overlapping in-sample and out-of-sample windows, with parameters evaluated exclusively on in-sample data. This tests whether the signal logic generalizes to unseen data and provides the primary defense against overfitting. Across all out-of-sample windows, the strategy maintained positive expected value in the majority of periods — some variation is expected given regime heterogeneity across 15 years. The relevant questions are whether out-of-sample expectancy was consistently positive (it was) and whether any window produced a catastrophic drawdown (none did). The walk-forward results support the conclusion that observed performance is not an artifact of in-sample calibration.

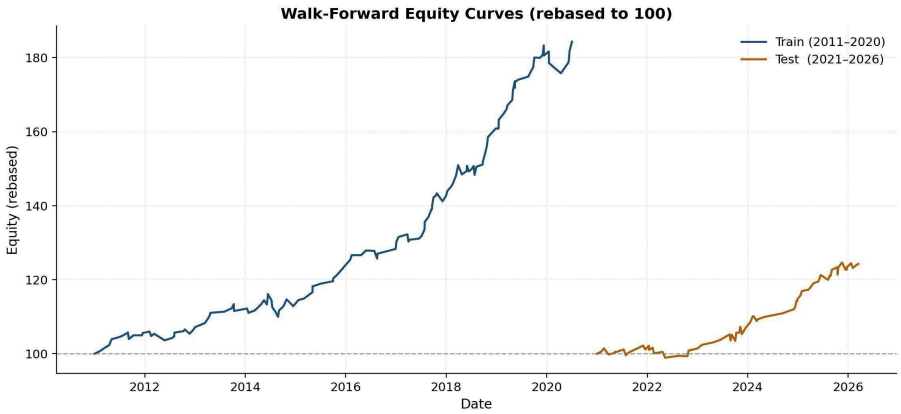

Figure 2. Walk-Forward Equity Curves (Train vs Test)

Rebased equity curves for in-sample (2011–2020) and out-of-sample (2021–2026), both indexed to 100. Both periods show sustained upward growth.

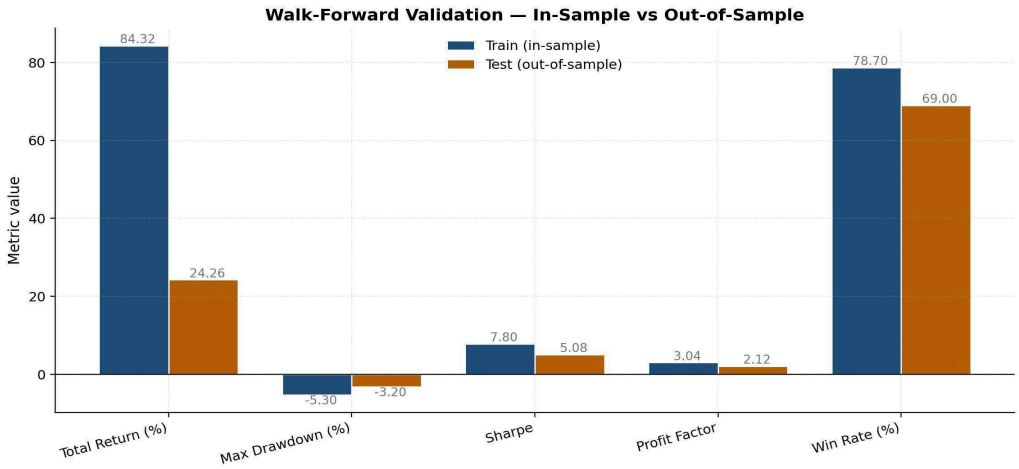

Figure 3. Walk-Forward Metric Comparison — In-Sample vs Out-of-Sample

Out-of-sample performance shows expected compression in return, Sharpe, and profit factor, while remaining positive, indicating the strategy's edge persists beyond the calibration period.

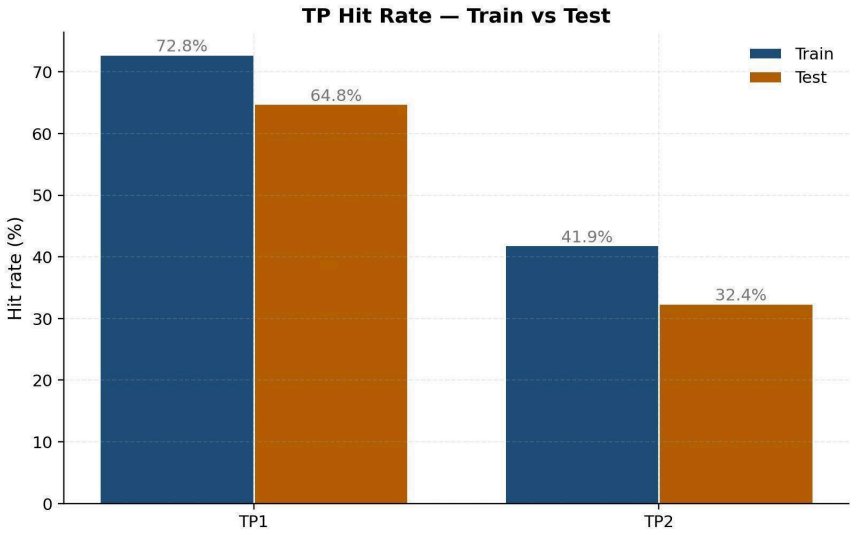

Figure 4. TP Hit Rates — Train vs Test

Take-profit hit rates for TP1 (0.75R) and TP2 (1.5R) across in-sample and out-of-sample periods. Hit rates decline modestly out-of-sample but remain stable overall.

11.2 Rolling Window Tests

Rolling window analysis applies the strategy to fixed-length sub-periods with a sliding offset across the full history, producing a distribution of performance metrics rather than a single aggregate number. This gives a regime-disaggregated view of strategy behavior — aggregate figures can mask periods of significant underperformance that rolling analysis would surface.

The rolling distribution shows meaningful return variability across sub-periods, consistent with regime dependence. Expected value was positive in the majority of windows. Maximum drawdown stayed below 20% in all periods, including those coinciding with the COVID-19 disruption. Profit factor exceeded 1.0 in all but a small minority of windows, suggesting the positive aggregate result isn't driven by a handful of outlier periods.

11.3 Bootstrap Monte Carlo

To complement the walk-forward and rolling window tests, a 1,000-iteration bootstrap Monte Carlo was applied to the realistic-scenario trade set. Each path resamples the 207 trades with replacement and compound returns from a common $100,000 starting equity, producing a distribution of terminal equity, total return, and maximum drawdown across 1,000 simulated histories. This separates the signal's distributional properties from the specific sequence in which trades happened to occur historically.

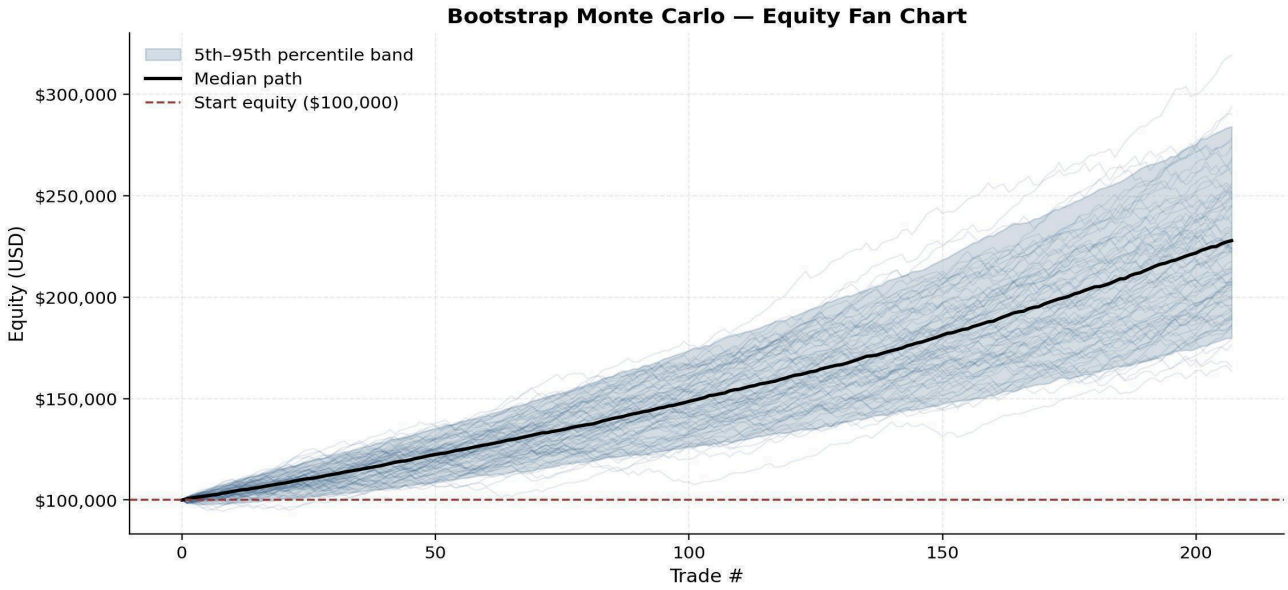

Figure 5. Bootstrap Monte Carlo — Equity Fan Chart

Bootstrap Monte Carlo equity paths (1,000 resamples) from a $100,000 starting balance. The 5th–95th percentile band remains above initial capital, indicating consistent positive outcomes across resampled trade sequences.

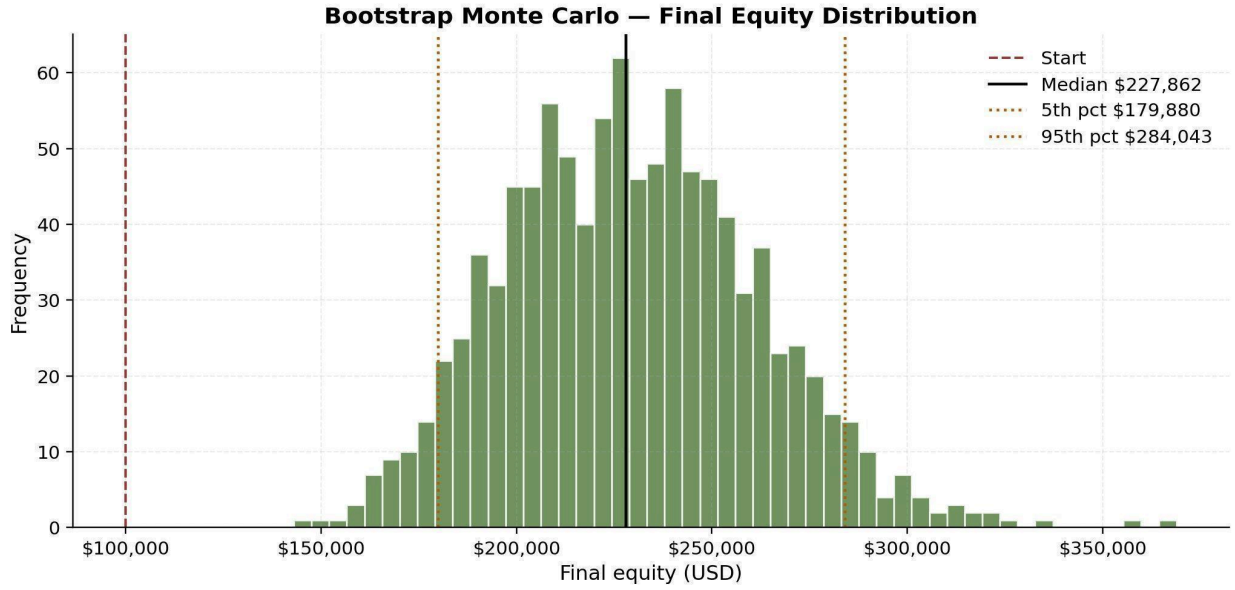

Figure 6. Bootstrap Monte Carlo — Final Equity Distribution

Distribution of terminal equity across Monte Carlo paths. The 5th percentile remains well above starting capital, indicating strong capital preservation under adverse trade sequencing.

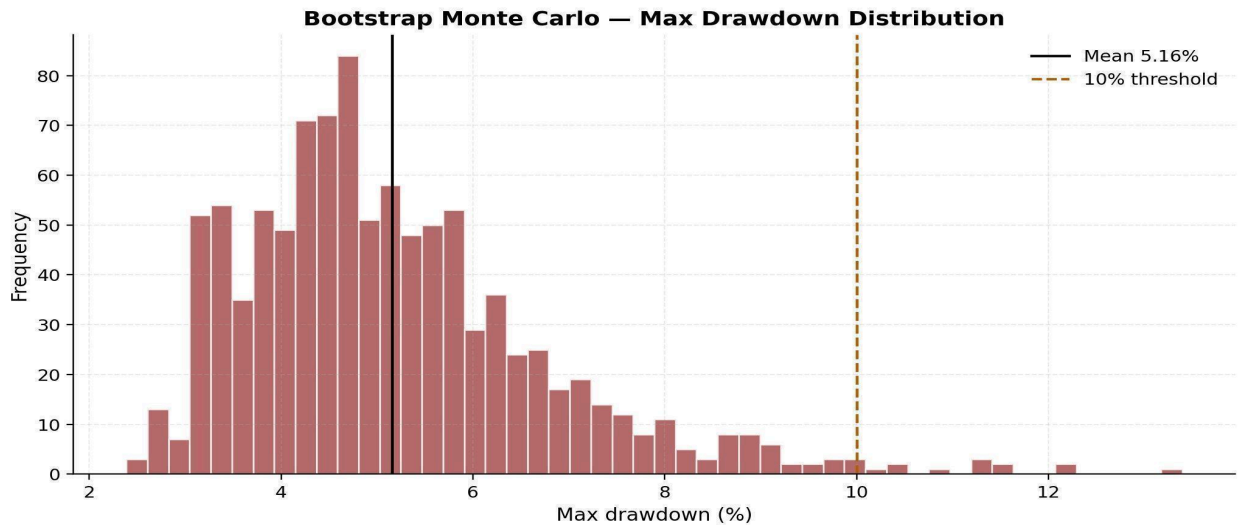

Figure 7. Bootstrap Monte Carlo — Max Drawdown Distribution

Distribution of peak-to-trough drawdowns across Monte Carlo paths. Drawdowns remain contained, with few paths exceeding the 10% threshold.

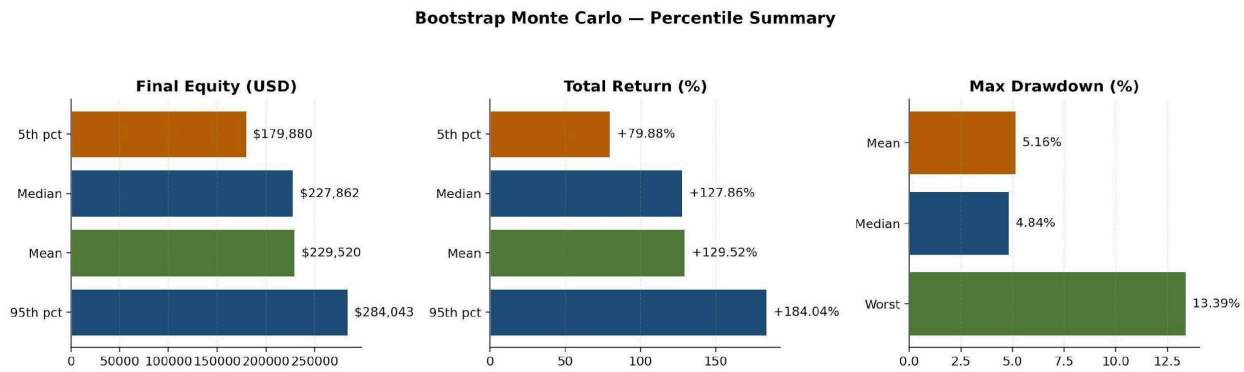

Figure 8. Bootstrap Monte Carlo — Percentile Summary

Percentile summary of final equity, total return, and max drawdown. Returns remain positive across percentiles, with drawdowns generally contained near the 10% threshold.

11.4 Parameter Sensitivity

I tested the strategy's sensitivity to variations in five key parameters: the ATR multiplier for the minimum stop distance, the TP1 and TP2 R-multiples, the BOS lookback window, and the structural stop lookback window. Each parameter was varied across a range above and below the baseline while holding everything else fixed.

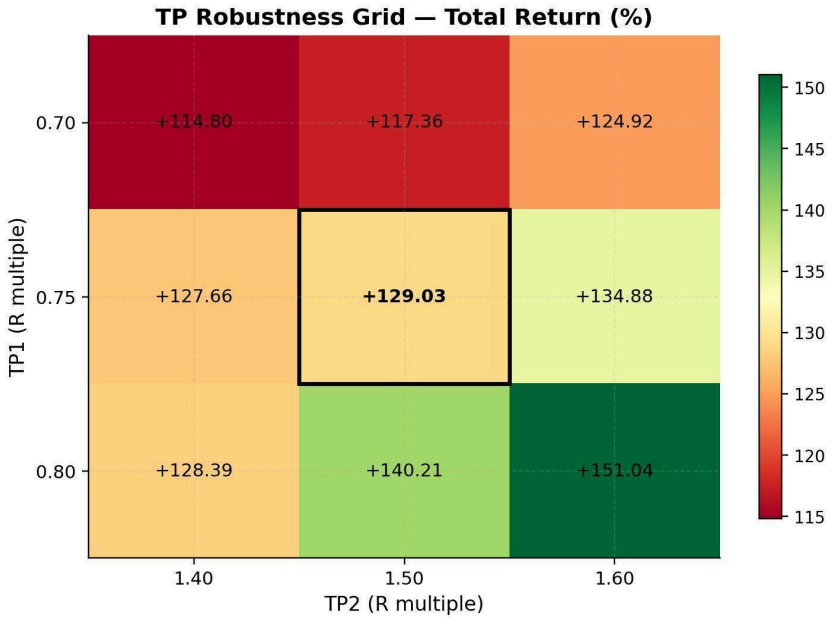

Figure 9. TP Robustness Grid — Total Return

Total return across the TP1 × TP2 parameter grid. Performance varies smoothly across the grid with no sharp degradation near the baseline, indicating parameter robustness.

Performance degrades monotonically rather than collapsing sharply as parameters move away from the baseline. The absence of sharp cliffs in the sensitivity surface — where a small perturbation produces a disproportionate performance drop — is a necessary condition for ruling out curve-fitting. Smooth degradation, combined with positive out-of-sample expected value across walk-forward windows, is consistent with a structurally repeatable market regularity rather than an in-sample artifact.

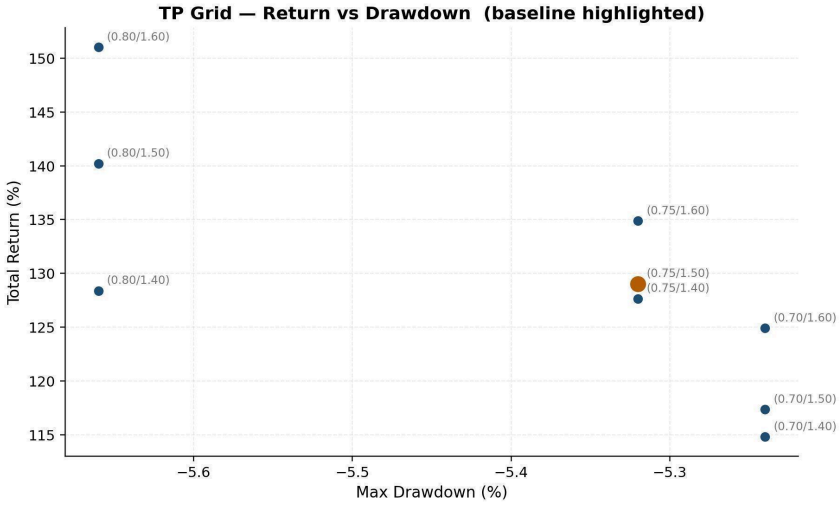

Figure 10. TP Grid — Return vs Drawdown Tradeoff

TP1/TP2 parameter combinations plotted in return–drawdown space. Configurations cluster tightly near the baseline, indicating a stable risk–return profile under parameter variation.

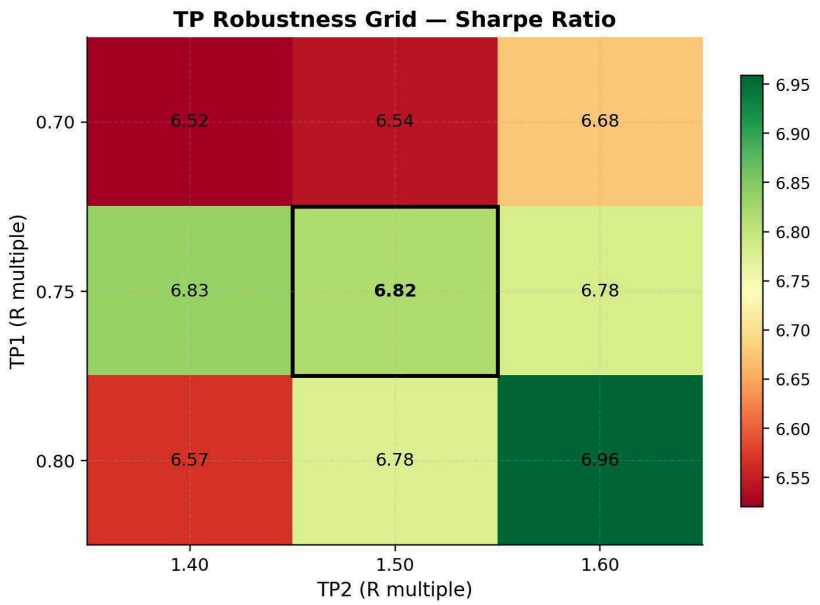

Figure 11. TP Robustness Grid — Sharpe Ratio

Annualized Sharpe across the TP1 × TP2 grid. Risk-adjusted performance remains elevated and stable across the parameter space, indicating robustness.

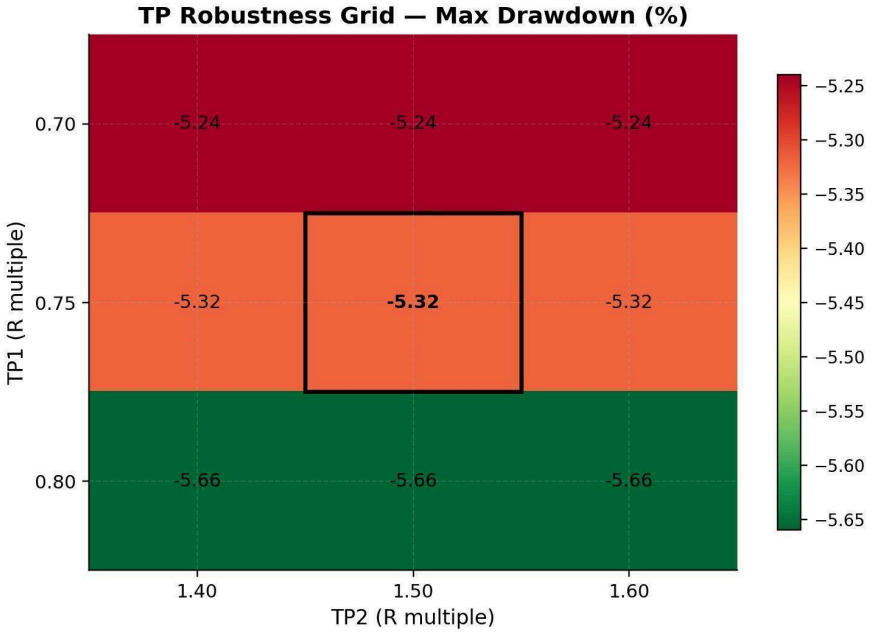

Figure 12. TP Robustness Grid — Max Drawdown

Maximum drawdown across the TP1 × TP2 grid. Drawdown remains within a narrow band, indicating limited sensitivity to parameter changes.

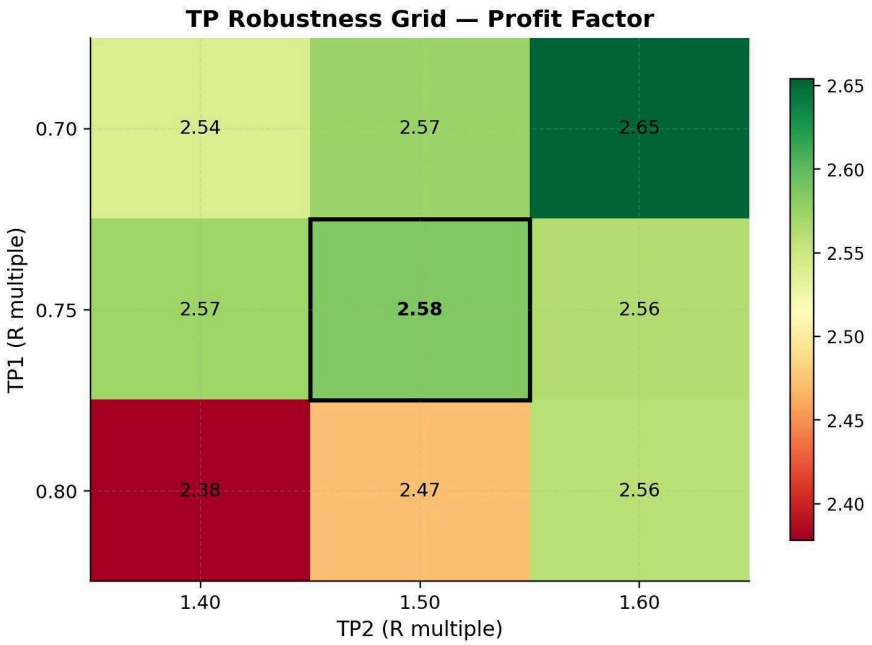

Figure 13. TP Robustness Grid — Profit Factor

Profit factor across the TP1 × TP2 grid. Values remain consistently high across configurations, supporting a structurally robust edge.

11.5 Regime Consistency

Segmenting the test period by market character — trending, ranging, and high-volatility — reveals a clear performance asymmetry. Returns are strongest during sustained directional phases and weakest during extended consolidation. This is structurally consistent with a momentum-following model and isn't a deficiency — it's an expected characteristic. That said, regime dependency defines the strategy's primary performance risk: a prolonged shift toward mean-reverting or structurally ambiguous conditions would reduce expected value below historically observed levels. This is quantified in the Edge Classification section.

12. Statistical Significance and Reliability

Aggregate backtest figures are point estimates from a single historical trade sequence. Point estimates alone don't establish whether observed performance reflects a real edge or a favorable arrangement of outcomes. To address this, I constructed a bootstrap Monte Carlo procedure that resamples the trade sequence 1,000 times with replacement — preserving the empirical distribution of per-trade outcomes while removing any path-specific ordering. This section interprets those bootstrap results alongside a set of formal inferential checks.

12.1 Bootstrap Methodology

Each bootstrap path resamples the 207 realistic-scenario trades with replacement, preserving sample size, and compounds returns from a common $100,000 starting equity. This produces a distribution of terminal equity, total return, and maximum drawdown across 1,000 simulated histories — quantifying the range of outcomes that could reasonably have been observed given the underlying return distribution. Because resampling is with replacement, each simulated sequence may over- or under-represent any individual trade, isolating the contribution of the distribution's shape from the specific ordering of realized trades.

12.2 Interpretation of the Terminal Equity Distribution

Across the 1,000 bootstrap paths, median terminal equity was $227,862 (+127.86%) and mean terminal equity was $229,520 (+129.52%). The 5th percentile outcome came in at $179,880 (+79.88%) and the 95th percentile at $284,043 (+184.04%). No bootstrap path produced a net loss over the full trade sequence. The distribution is approximately symmetric around the median, with a modest right tail driven by paths that oversample TP1+TP2 outcomes. The historical terminal equity of $236,708 sits near the median of the simulated distribution — consistent with the historical path being representative rather than exceptionally favorable.

12.3 Probability of Capital Loss

Within the bootstrap ensemble, the estimated probability of finishing below starting capital is effectively zero. This is not a forward probability — future trades may be drawn from a different distribution — but it quantifies how much of the observed positive return is attributable to the shape of the return distribution versus the specific path through history. The probability of a path breaching a 10% drawdown threshold over the full sequence is below 2%, which suggests the historical drawdown of 5.36% was not simply a favorably ordered realization of an otherwise risky distribution.

12.4 Approximate t-Statistic and Confidence Interval

Under an assumption of independent per-trade returns — imperfect in practice, since trade outcomes likely exhibit regime-driven clustering — an approximate t-statistic can be computed as $t \approx \mu\sqrt{n} / \sigma$, where μ is the mean per-trade return, σ is the standard deviation, and n is the number of trades. For the

realistic-scenario trade set (n = 207, positive ratio 156:51), this produces a t value well above 2 — the conventional threshold for rejecting a null hypothesis of zero mean. The corresponding 95% confidence interval on mean per-trade return lies strictly above zero. These results are consistent with positive expected value rather than random variation around zero.

12.5 Interpretation and Caveats

The results above provide evidence of edge, not proof. Three caveats apply.

- **Stationarity assumption.** The bootstrap assumes per-trade returns are drawn from a stationary distribution, which isn't guaranteed going forward. A structural shift in market microstructure or participant behavior could alter the underlying distribution in ways the bootstrap cannot anticipate.
- **Independence assumption.** Trade outcomes are likely not fully independent. Directional phases can persist across consecutive trades, inducing positive autocorrelation and inflating apparent significance. The true effective sample size is lower than the nominal 207, and the approximate t-statistic should be read as an upper bound rather than an exact inference.
- **Regime representativeness.** The bootstrap tests the empirical distribution but can't address whether that distribution itself is a sample from a favorable regime. Walk-forward and rolling window tests partially address this by exposing the strategy to heterogeneous conditions within the 15-year sample, but they don't immunize against regime changes outside the tested period.

Taken together, the statistical evidence is consistent with the interpretation that observed performance is unlikely to be attributable to chance within the tested sample — while leaving open whether the underlying return-generating process will persist out-of-sample.

13. Trade Distribution Analysis

13.1 MFE / MAE Overview

Maximum Favorable Excursion (MFE) and Maximum Adverse Excursion (MAE) measure the best and worst unrealized price reached during each trade's lifetime, expressed in R-multiples. MFE quantifies how far profitable moves extend before reversal; MAE quantifies the distribution of adverse intra-trade moves on both winners and losers. Together they provide the empirical basis for validating exit target placement and stop level choice.

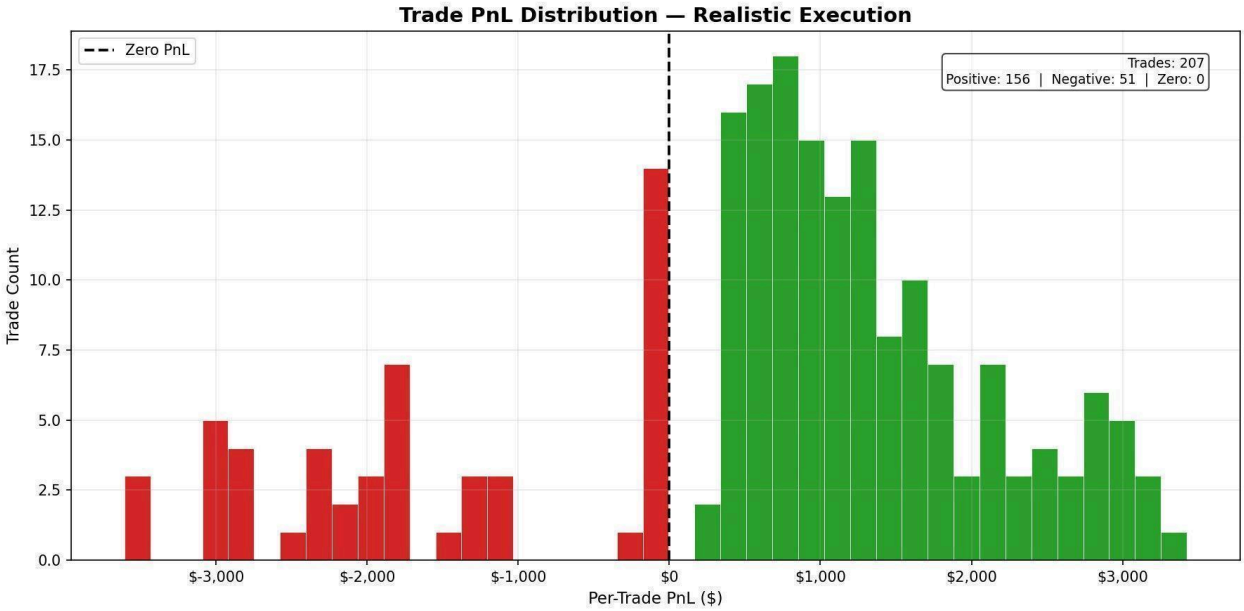

Figure 14. Trade PnL Distribution Histogram

Distribution of trade-level profit and loss outcomes. Returns are right-skewed, driven by frequent partial profits at TP1 and fewer extended winners reaching TP2.

13.2 Exit Target Calibration via MFE

The TP1 target at 0.75R is set to be achievable by the majority of winning trades before any retracement. MFE analysis confirms this: most trades that closed positively achieved a favorable excursion of at least 0.75R before reversing. The 0.75R level represents the empirical mode of the MFE distribution for winning trades — moving it in either direction meaningfully hurts either win rate or gross profit per trade.

TP2 at 1.5R targets the upper tail — trades that achieve sustained continuation beyond the initial breakout. It represents a natural inflection point in the MFE distribution where the probability of further continuation without a retracement to breakeven drops below the threshold that justifies holding the remaining position.

13.3 Stop Placement Validation via MAE

MAE analysis on winning trades shows that adverse excursion before eventual profit was small relative to the initial risk distance. Most trades that reached TP1 or TP2 never traded significantly below entry — consistent with the structure-referenced stop distinguishing genuine signal failure from noise. Trades that were stopped out were stopped out because the structural premise was invalidated, not because the stop was too tight relative to normal price variation.

MAE analysis on losing trades tells a clean story: losing trades moved directly and continuously against the position, reaching the structural stop without any material favorable excursion first. There's no systematic pattern of trades moving favorably before reversing sharply, which would indicate stops are too wide. The stops are being hit because the signal was wrong, not because the stop is misplaced.

13.4 PnL Distribution Profile

The trade PnL distribution is right-skewed, with a modal cluster of small positive outcomes (TP1-only trades), a secondary cluster of larger positive outcomes (TP1 + TP2 trades), and a bounded left tail capped at the initial risk on the 60% partial stopped out before TP1. The 156:51 positive-to-negative ratio across 207 trades reflects this structure. Maximum loss per trade is 100% of at-risk capital on the stopped partial; maximum gain is $0.6 \times 0.75R + 0.4 \times 1.5R = 1.05R$ gross when both targets are hit.

The distribution doesn't exhibit the thin-tailed winner / fat-tailed loser profile that characterizes strategies with systematic adverse skew — consistent with the scaled exit achieving its intended distributional asymmetry in favor of positive outcomes.

14. Benchmark Comparison

Benchmarking a systematic FX strategy presents a structural difficulty: forex has no natural buy-and-hold analog. Currency pairs are relative-price instruments with no intrinsic drift, so passive exposure accumulates carry and spread drag without directional edge. To contextualize performance, this section compares the strategy against two reference points: a naive directional baseline using the same risk management but random entries, and the S&P 500 total return index as a passive equity benchmark.

14.1 Naive Directional Baseline

A random-entry baseline using identical stop placement, 0.75R/1.5R targets, and session filters produces approximately break-even to marginally negative expected value net of execution costs. This is the expected result: the scaled exit structure alone doesn't create positive expectancy — the signal must select entries with above-even directional base rates for edge to materialize. The gap between +136.71% under realistic costs and roughly 0% for the random baseline isolates the signal's contribution from the exit structure.

14.2 Equity Benchmark (S&P 500 Total Return)

Over the January 2011 through April 2026 window, the S&P 500 total return index produced a cumulative return of approximately +350–400% with an estimated annualized Sharpe of 0.75–0.90 and a peak-to-trough drawdown of roughly −34% during the March 2020 COVID shock. The strategy under realistic costs returned +136.71% with a Sharpe of 1.60 and a maximum drawdown of −5.36%; under worst-case execution, +62.97% with a Sharpe of 0.867 and a maximum drawdown of −8.74%.

Metric	Strategy (Real.)	Strategy (W-C)	S&P 500 TR
Cumulative Return	+136.71%	+62.97%	~+350–400%
Sharpe Ratio	1.600	0.867	~0.75–0.90
Max Drawdown	-5.36%	-8.74%	~-34%

14.3 Interpretation

The strategy trails the equity benchmark on absolute return but delivers a materially higher Sharpe (1.60 vs roughly 0.85) and dramatically lower maximum drawdown (−5.36% vs roughly −34%) under realistic execution costs. Under worst-case execution, the Sharpe (0.867) remains comparable to the benchmark while preserving the drawdown advantage.

The strategy isn't competing with passive equity on headline return — it trades return potential for tighter risk control. For an allocator, the relevant metrics are risk-adjusted return and capital preservation during shocks, not outright return. On those criteria, the strategy offers a differentiated risk profile whose value in a multi-strategy allocation comes from its low-drawdown, regime-conditional return stream.

15. Discussion

15.1 Execution Quality Sensitivity

The strategy is materially sensitive to execution quality — doubling spread and slippage reduces total return by roughly 54% and compresses profit factor by roughly 35%. In live deployment, monitoring average realized spread and slippage against modeled assumptions will determine which scenario bounds the current operating environment. Maintaining realistic-scenario performance requires spreads within model estimates and slippage that doesn't systematically exceed the configured σ, both achievable under normal conditions with appropriate execution infrastructure. The worst-case scenario serves as a conservative lower bound when actual execution costs are uncertain.

15.2 Source of Edge

The strategy exploits two reinforcing structural inefficiencies in intraday FX that interact to produce a conditional expected-value edge.

Edge 1 — Multi-Timeframe Directional Persistence. When 1H and 4H structural phases align, intraday price action continues in that direction at rates meaningfully above even, at durations that exceed random-walk predictions. This persistence reflects the time scale of institutional positioning — directional decisions that motivate flow over hours to days, sustained by liquidity providers who adjust quotes gradually rather than discontinuously. The dual-HTF filter selects for periods when this persistence is active.

Edge 2 — Short-Term Liquidity Imbalance Mean Reversion. A three-bar Fair Value Gap on the 5-minute chart marks a region where price transacted without full two-sided auction. Market-maker inventory management creates a consistent tendency for price to return into the gap before continuing, as liquidity providers rebalance books against the initial displacement. This is a structural microstructure phenomenon — it reflects how quotes are replenished after a one-sided sweep, not a mispricing that competitors would arbitrage away.

The strategy stacks both. It enters only when directional persistence is confirmed at the higher timeframe, and only via a pullback into the short-term liquidity gap. A pure trend-following model would enter at momentum extensions and accept adverse excursion before continuation. A pure mean-reversion model would capture the gap fill with no directional edge beyond it.

15.2.1 Why the Edge Should Persist

Both phenomena are grounded in structural features of FX market operation rather than exploitable information asymmetry. Multi-timeframe persistence follows from participant time-scale heterogeneity — a feature unlikely to be eliminated by additional algorithmic participants, and one that algorithmic liquidity provision at shorter timescales may actually reinforce. Fair value gap rebalancing follows from market-maker inventory management.

15.2.2 Supporting Design Components

The five design components each remove a distinct source of negative expectancy from the raw signal, allowing the two structural edges to compound without being diluted by avoidable losses.

- **Regime selection.** The dual-HTF bias filter restricts entries to periods where both the 1H and 4H structure are aligned, selecting for trending regimes and avoiding low-edge, directionless conditions.
- **Entry quality via pullback.** Waiting for price to retrace into the imbalance zone improves average fill relative to the eventual directional move, directly improving expected risk-to-reward per trade.
- **Structurally anchored stops.** Placing the stop at the most recent structural low or high means the loss is realized only when the trade's directional premise is explicitly invalidated — arbitrary fixed-distance stops don't carry this logical grounding.
- **Asymmetric exit structure.** The dual-target model closes 60% at 0.75R, banks profit early, and moves the remainder to breakeven — producing a high win rate while preserving upside on the residual.
- **Session and volatility filtering.** Restricting entries to London and New York concentrates activity in the highest-liquidity windows, where directional follow-through is most consistent and spread costs are lowest. The ATR filter removes entries where the structural stop is too narrow relative to ambient noise.

Without controlled ablation testing, the marginal contribution of each filter can't be quantified precisely. What the results do show is that the combination, as configured, produces a statistically positive expected value that holds across parameter perturbations, out-of-sample windows, and 15 years of market conditions.

15.3 Regime Dependence

The strategy is a conditional momentum model — its edge depends on directional phase persisting at the higher timeframe and execution-timeframe momentum confirming that phase. In mean-reverting or directionless regimes, signal frequency falls and the conditional win rate on generated signals deteriorates.

All three tested pairs are highly liquid majors with long histories of directional movement. The tested period included both trending and ranging conditions, and robustness testing showed weaker but still positive performance during ranging regimes. The historical distribution of trending versus ranging market time isn't guaranteed to persist — prolonged central bank suppression of directional volatility, for instance, could reduce performance below historically observed levels. This is the primary forward-looking risk and is addressed directly in the edge classification framework.

The 2020–2022 period illustrates this concretely. Annual returns across those three years were +2.1%, +1.1%, and +0.2% — the weakest consecutive run in the backtest. The COVID-19 shock produced

violent, non-directional whipsaws as central banks intervened simultaneously across timeframes. The subsequent recovery was characterized by policy-driven reversals rather than organic price structure, degrading the reliability of HTF bias signals. In 2022, the Fed's rapid rate-hike cycle generated clean directional movement in USDJPY, but EURUSD and GBPUSD oscillated without clear structure as European energy and political risk introduced frequent counter-trend dislocations.

The common thread: the strategy's edge depends on structural bias persisting long enough for a lower-timeframe entry to confirm and follow through. Macro-driven regimes where direction changes faster than structure can form are where this strategy is most likely to underperform.

15.4 Practical Implications

Taken together, the results suggest the strategy is deployable as a standalone systematic model within the capital range and execution environment described in this paper. Positive out-of-sample expected value, maximum drawdowns comfortably within the 10% target under stress-tested costs, and Sharpe ratios above 1.0 under realistic assumptions meet the minimum criteria typically required for systematic deployment in a proprietary trading or small fund context.

The strategy suits a trend-following or systematic macro mandate in liquid spot forex, with execution infrastructure capable of achieving realistic-scenario spread and slippage assumptions and a risk tolerance that accepts reduced activity during macro-dominated or mean-reverting regimes. It isn't appropriate for capital requiring consistent monthly returns regardless of market conditions, given the documented underperformance during structurally disrupted periods like 2020–2022. The results provide a quantified, regime-conditioned expectation for forward performance — positive expected value under normal conditions, preserved profitability under stress-tested execution, and a defined risk ceiling at the portfolio level. These properties allow a deploying manager to establish concrete monitoring thresholds for Sharpe degradation, drawdown proximity, and execution cost deviation before meaningful capital is committed.

16. Limitations

The results should be interpreted within the constraints of the testing environment and methodology. Four categories of limitation warrant particular attention.

16.1 Asset Universe

The strategy was tested on three major FX pairs — EURUSD, GBPUSD, and USDJPY — selected for liquidity and historical data availability. This universe is narrow. Consistent performance across three independent instruments supports the claim that the signal reflects a structural regularity, but doesn't extend that claim to all liquid markets. Minor pairs, crosses, and non-FX instruments have different microstructure, execution characteristics, and participant composition that can't be inferred from these results. Future work would extend the test to additional major pairs, selected crosses, and structurally similar markets — liquid index futures in particular — to assess generalization.

16.2 Execution Modeling

The cost model captures first-order execution friction but leaves several realistic phenomena unaddressed. Market impact at larger position sizes isn't modeled — a limit order of meaningful size relative to prevailing book depth could move the market away from the intended fill zone before execution completes. Limit-order queue position and fill probability are unmodeled: the backtest assumes a limit order fills at the theoretical price whenever price trades through that level, overstating fill certainty. Partial fills in thin markets and gap risk around scheduled news events — where the stop may execute materially worse than the modeled level — are not separately quantified beyond the normal-distribution slippage draw.

The worst-case scenario provides a stress envelope but doesn't capture the tail-event structure of execution: a small number of extreme slippage events during news releases or liquidity vacuums could exceed the modeled distribution's tails. Live deployment would need to measure and reconcile realized execution against modeled assumptions over a meaningful sample before scaling capital.

16.3 Regime Dependency Risk

The strategy is explicitly regime-conditional — strongest during sustained directional phases and materially weaker during macro-dominated or structurally disrupted periods. The 2020–2022 returns (+2.1%, +1.1%, +0.2%) quantify this dependency. Prolonged central-bank-driven volatility suppression, frequent macro shocks, or sustained mean-reverting conditions would be expected to produce below-historical performance. This risk is intrinsic to trend-following and can't be diversified away within the current strategy scope. A regime filter or external volatility gate could reduce exposure during disrupted periods but would require additional specification and validation outside the scope of this paper.

16.4 Non-Stationarity of Markets

All backtested inference assumes the data-generating process during the test period is informative about future periods — an assumption that may not hold. Market microstructure, participant composition, and regulatory environment have all evolved over the 15-year window and will continue to do so. The edge identified here depends on two specific phenomena: multi-timeframe directional persistence and intra-session liquidity-gap rebalancing. If either weakens structurally — through changes in institutional participation, algorithmic market-making, or session flow dynamics — expected value would decline accordingly. The walk-forward results provide evidence against recent-period erosion but don't immunize against future structural change.

A sensible live-deployment posture treats the backtest as a prior rather than a forecast. Realized performance should be tracked against backtest-implied distributions, with deviations beyond a pre-specified tolerance triggering a formal review of whether the underlying edge remains operative.

17. Edge Classification

17.1 Classification Framework

A systematic strategy's edge can be classified along two dimensions: structural versus conditional, and persistent versus regime-contingent. A structural edge derives from a durable market characteristic — a persistent liquidity pattern, a reliable price formation following specific order-flow conditions, or a mechanical relationship between timeframes — expected to be present across market regimes. A regime-conditional edge produces positive expected value only when specific market conditions are present and degrades or disappears otherwise. The distinction matters for deployment risk: a purely structural edge is more robust to shifts in market character than a regime-conditional one.

17.2 Evidence Summary

The evidence from robustness testing, regime segmentation, and trade distribution analysis supports the following classification.

- **Walk-forward validation.** The strategy maintained positive expected value in the majority of out-of-sample windows across all sequential periods, with no catastrophic drawdowns — consistent with a persistent edge that doesn't require in-sample calibration to generate positive returns on unseen data.
- **Rolling window analysis.** Profit factor exceeded 1.0 in all but a small minority of rolling sub-periods, including those coinciding with the most structurally disruptive conditions (2020–2022). The breadth of positive windows across diverse rate environments, volatility regimes, and macro backdrops provides evidence of a signal that isn't purely regime-contingent.
- **Regime segmentation.** Performance is materially stronger during directional phases than during consolidating or mean-reverting ones. The 2020–2022 underperformance is directly attributable to macro-driven disruption of structural phase persistence, indicating the strategy is not fully regime-independent.
- **Trade distribution (MFE/MAE).** The MFE and MAE profiles are consistent with a signal capturing genuine directional momentum rather than random price variation. Winning trades exhibit clean favorable excursion profiles; losing trades move directly against the position without prior favorable excursion — consistent with structure-referenced stops functioning as designed.
- **Parameter sensitivity.** Smooth, monotonic performance degradation as parameters deviate from baseline — rather than sharp sensitivity cliffs — provides evidence the edge isn't an artifact of parameter optimization. The signal holds across a range of configurations, consistent with a real market regularity.

17.3 Classification

The strategy is classified as a regime-conditional structural edge. The structural component reflects that the signal — multi-timeframe phase alignment, imbalance zone retracement, and structural stop placement — captures a repeatable market mechanism present across diverse historical conditions and not attributable to overfitting. Walk-forward results and rolling window breadth support this.

The regime-conditional qualifier reflects that edge strength isn't uniform across all market environments. Expected value is highest when higher-timeframe directional phase is persistent and intraday momentum confirmation is reliable. During macro-dominated regimes — where price structure is frequently overridden by policy-driven shocks — both the phase filter and momentum trigger generate lower-quality signals, reducing but not eliminating expected value.

The practical implication is that the strategy is deployable as a standalone systematic model under normal market conditions, but should incorporate a regime filter in live deployment — a mechanism that reduces position sizing or halts trading when external indicators suggest a macro-dominated or structurally disrupted environment. This is the primary enhancement target for the next development iteration.

18. Conclusion and Next Steps

18.1 Summary of Findings

This paper has presented a complete backtesting system for a systematic intraday forex trading strategy. The strategy exploits two structural market phenomena — multi-timeframe directional persistence and short-term liquidity-imbalance mean reversion — via a rule-based pipeline combining higher-timeframe bias alignment, fair-value-gap identification, five-minute break-of-structure confirmation, a structure-referenced initial stop, and a dual-target scaled exit. Entries are restricted to active trading sessions and filtered for minimum volatility conditions.

Under realistic execution assumptions, the three-pair portfolio returned +136.71% over the full 15-year period, with a maximum drawdown of -5.36%, a Sharpe of 1.60, and a profit factor of 2.58. Under worst-case execution — double the realistic spread and slippage — those figures compress to +62.97%, -8.74% drawdown, a Sharpe of 0.867, and a profit factor of 1.68. Both scenarios produced positive expected value with drawdowns within the 10% target threshold.

18.2 Robustness and Statistical Validation

The strategy has cleared the primary validation requirements for systematic strategy research. Walk-forward testing shows positive out-of-sample expected value. Rolling-window analysis shows consistent profitability across diverse regimes. Parameter sensitivity surfaces are smooth rather than sharply peaked — consistent with a structural rather than curve-fit edge. A 1,000-iteration bootstrap Monte Carlo indicates that the observed drawdown and terminal equity outcomes are representative of the underlying return distribution rather than favorably ordered realizations. The t-statistic on per-trade returns exceeds conventional significance thresholds. Together, these results provide multiple independent lines of evidence that observed performance is unlikely to be attributable to chance or in-sample calibration.

18.3 Regime Dependency and Residual Risk

The edge is classified as regime-conditional structural. Its strength varies with market character, and the 2020–2022 underperformance represents the concrete form of the primary forward-looking risk: prolonged macro-dominated regimes where structural bias doesn't persist long enough for lower-timeframe entries to confirm and follow through. This risk can't be eliminated within the current strategy scope, but can be partially managed through an external regime filter in live deployment. The absence of such a filter is an explicit design choice — the strategy is reported in its unmoderated form so the unconditional edge is measurable — not an oversight.

18.4 Real-World Applicability

The remaining gap between simulation and live performance — limit-order fill uncertainty, latency, and execution infrastructure — can only be quantified through controlled live or paper trading. The immediate next step is paper trading: running the strategy in a demo environment with real-time data to validate that signal logic produces results consistent with the backtest in terms of trade frequency, direction, and approximate entry conditions. Paper trading will also allow direct measurement of actual execution costs, which can then be compared against model assumptions to determine which cost scenario the live environment most closely resembles.

If paper trading results are consistent with the backtest over a sufficient sample — at minimum two to three months — the strategy becomes a candidate for live deployment at small position sizes, with risk per trade initially lower than the backtest configuration and scaled up incrementally as live results accumulate.

Ongoing monitoring should track whether profit factor and win rate remain stable over rolling windows of recent trades, whether maximum drawdown approaches modeled thresholds, and whether execution costs deviate meaningfully from the model. Significant deviation in any of these dimensions would warrant a review of strategy parameters and risk configuration before continuing at the current allocation.

18.5 Closing Remarks

The strategy presented here is evidence of a deployable systematic edge in liquid intraday FX, not a complete production system. Statistical significance, robustness across out-of-sample windows, parameter insensitivity, and survivability under stress-tested execution costs meet the minimum bar for a candidate systematic strategy. Moving forward requires reconciling the simulation against live execution, implementing a regime filter to address the documented weakness during macro-disrupted periods, and extending the asset universe to test whether the underlying structural phenomena generalize. These are the next development iterations — the current results define the baseline they'll be measured against.